

Ques : 1. List all students.

Ans :

```
SELECT *  
FROM Students;
```

Results

Messages

	StudentID	FirstName	LastName	Gender	DateOfBirth	Email	EnrollmentDate
1	1	Alice	Johnson	F	2000-01-15	alice.j@example.com	2021-08-01
2	2	Bob	Smith	M	1999-07-23	bob.s@example.com	2021-08-01
3	3	Charlie	Lee	M	2001-02-17	charlie.l@example.com	2022-01-10
4	4	Diana	Brown	F	1998-11-30	diana.b@example.com	2020-09-01
5	5	Ethan	Hunt	M	1997-03-25	ethan.h@example.com	2020-09-01
6	6	Fatima	Ali	F	2002-06-19	fatima.a@example.com	2022-08-01
7	7	George	Williams	M	1998-12-01	george.w@example.com	2021-01-05
8	8	Hannah	Kim	F	1999-11-12	hannah.k@example.com	2022-01-01

Ques : 2. Show students' full names and emails.

Ans :

```
SELECT  
  StudentID,  
  CONCAT(FirstName, ' ', LastName) AS FullName,  
  Email  
FROM Students;
```

Results		Messages	
	StudentID	FullName	Email
1	1	Alice Johnson	alice.j@example.com
2	2	Bob Smith	bob.s@example.com
3	3	Charlie Lee	charlie.l@example.com
4	4	Diana Brown	diana.b@example.com
5	5	Ethan Hunt	ethan.h@example.com
6	6	Fatima Ali	fatima.a@example.com
7	7	George Williams	george.w@example.com
8	8	Hannah Kim	hannah.k@example.com

Ques : 3. Find all courses worth more than 3 credits.

Ans :

```
SELECT *  
FROM Courses  
WHERE Credits > 3;
```

 Results  Messages

	CourseID	CourseName	CourseCode	Credits	DepartmentID
1	2	Programming in C#	CS102	4	1
2	5	Classical Mechanics	P101	4	3

Ques : 4. Get students who enrolled after January 1, 2022.

Ans :

```
SELECT *  
FROM Students  
WHERE EnrollmentDate > '2022-01-01';
```

 Results  Messages

	StudentID	FirstName	LastName	Gender	DateOfBirth	Email	EnrollmentDate
1	3	Charlie	Lee	M	2001-02-17	charlie.l@example.com	2022-01-10
2	6	Fatima	Ali	F	2002-06-19	fatima.a@example.com	2022-08-01

Ques : 5. Show all female students.

Ans :

```
SELECT *  
FROM Students  
WHERE Gender = 'F';
```

Results

Messages

	StudentID	FirstName	LastName	Gender	DateOfBirth	Email	EnrollmentDate
1	1	Alice	Johnson	F	2000-01-15	alice.j@example.com	2021-08-01
2	4	Diana	Brown	F	1998-11-30	diana.b@example.com	2020-09-01
3	6	Fatima	Ali	F	2002-06-19	fatima.a@example.com	2022-08-01
4	8	Hannah	Kim	F	1999-11-12	hannah.k@example.com	2022-01-01

Ques : 6. Order students by last name descending.

Ans :

```
SELECT *  
FROM Students  
ORDER BY LastName DESC, FirstName DESC;
```

Results

Messages

	StudentID	FirstName	LastName	Gender	DateOfBirth	Email	EnrollmentDate
1	7	George	Williams	M	1998-12-01	george.w@example.com	2021-01-05
2	2	Bob	Smith	M	1999-07-23	bob.s@example.com	2021-08-01
3	3	Charlie	Lee	M	2001-02-17	charlie.l@example.com	2022-01-10
4	8	Hannah	Kim	F	1999-11-12	hannah.k@example.com	2022-01-01
5	1	Alice	Johnson	F	2000-01-15	alice.j@example.com	2021-08-01
6	5	Ethan	Hunt	M	1997-03-25	ethan.h@example.com	2020-09-01
7	4	Diana	Brown	F	1998-11-30	diana.b@example.com	2020-09-01
8	6	Fatima	Ali	F	2002-06-19	fatima.a@example.com	2022-08-01

Ques : 7. Count total students.

Ans :

```
SELECT COUNT(*) AS TotalStudents  
FROM Students;
```

Results Messages	
	TotalStudents
1	8

Ques : 8. Count number of courses per department.

Ans :

```
SELECT  
  d.DepartmentID,  
  d.DepartmentName,  
  COUNT(c.CourseID) AS CourseCount  
FROM Departments d  
LEFT JOIN Courses c  
  ON c.DepartmentID = d.DepartmentID  
GROUP BY d.DepartmentID, d.DepartmentName  
ORDER BY d.DepartmentName;
```

Results Messages			
	DepartmentID	DepartmentName	CourseCount
1	4	Business Administration	2
2	1	Computer Science	3
3	2	Mathematics	2
4	3	Physics	1

Ques : 9. Show all courses with their department names.

Ans :

```
SELECT
c.CourseID,
c.CourseName,
c.CourseCode,
c.Credits,
d.DepartmentName
FROM Courses c
JOIN Departments d
ON d.DepartmentID = c.DepartmentID
ORDER BY d.DepartmentName, c.CourseName;
```

Results		Messages			
	CourseID	CourseName	CourseCode	Credits	DepartmentName
1	8	Business Analytics	B201	3	Business Administration
2	6	Marketing Basics	B101	3	Business Administration
3	7	Advanced SQL Queries	CS201	3	Computer Science
4	1	Database Systems	CS101	3	Computer Science
5	2	Programming in C#	CS102	4	Computer Science
6	3	Calculus I	M101	3	Mathematics
7	4	Linear Algebra	M102	3	Mathematics
8	5	Classical Mechanics	P101	4	Physics

Ques : 10. List all students and the courses they are enrolled in.

Ans :

```
SELECT
s.StudentID,
CONCAT(s.FirstName, ' ', s.LastName) AS StudentName,
c.CourseID,
c.CourseName,
c.CourseCode,
e.EnrollmentDate,
e.Grade
FROM Students s
LEFT JOIN Enrollments e
ON e.StudentID = s.StudentID
LEFT JOIN Courses c
ON c.CourseID = e.CourseID
ORDER BY s.StudentID, c.CourseName;
```

 Results  Messages

	StudentID	StudentName	CourseID	CourseName	CourseCode	EnrollmentDate	Grade
1	1	Alice Johnson	7	Advanced SQL Queries	CS201	2022-01-15	A
2	1	Alice Johnson	1	Database Systems	CS101	2021-08-10	A
3	1	Alice Johnson	2	Programming in C#	CS102	2021-08-10	B
4	2	Bob Smith	3	Calculus I	M101	2021-08-10	B
5	2	Bob Smith	1	Database Systems	CS101	2021-08-10	C
6	2	Bob Smith	4	Linear Algebra	M102	2022-01-10	A
7	3	Charlie Lee	8	Business Analytics	B201	2022-08-01	A
8	3	Charlie Lee	5	Classical Mechanics	P101	2022-01-20	B
9	3	Charlie Lee	2	Programming in C#	CS102	2022-01-20	A
10	4	Diana Brown	7	Advanced SQL Queries	CS201	2021-08-10	A
11	4	Diana Brown	6	Marketing Basics	B101	2020-09-15	B
12	5	Ethan Hunt	1	Database Systems	CS101	2020-09-10	A
13	5	Ethan Hunt	4	Linear Algebra	M102	2021-01-10	B
14	6	Fatima Ali	8	Business Analytics	B201	2022-08-15	B
15	6	Fatima Ali	2	Programming in C#	CS102	2022-08-15	A
16	7	George Williams	3	Calculus I	M101	2021-02-10	C
17	7	George Williams	5	Classical Mechanics	P101	2021-02-10	A
18	8	Hannah Kim	7	Advanced SQL Queries	CS201	2022-08-01	B
19	8	Hannah Kim	1	Database Systems	CS101	2022-01-10	A
20	8	Hannah Kim	6	Marketing Basics	B101	2022-01-10	A